Project Title:

Conversion of post-combustion gas from cement manufacturing to syngas by electrochemical CO_2 capture and reduction

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May monthly report

1. **Tasks list for April**

- CO2RR performance test of varied Ni-2D-NC mass loadings under higher applied current densities with a cell config. of IrO_2/AEM/Ni-2D-NC.
- Investigation of CO_2 partial pressure (15, 30, 50, and 100%) under 200mA/cm^2 and
- Fabrication and electrocatalytic studies of deposited IrO_2-TiPt based anode catalyst.

2. **Experimental Procedure**

Fabrication of IrO_x/TiPt-felt electrode
The IrO_x/TiPt-felt anode electrode was prepared via thermal deposition method as following step by step:

Etching process
1. Sonically washing TiPt felt with DI water and acetone.
2. Etching Ti/Pt felt in 10ml of 6M HCl at 90°C for 45 min.
3. Weighing 3x3 cm^2 TiPt felt.

Preparation of dip coating solution
4. 30 mg of $IrCl_6.xH_2O$ was added into 10mL of IPA with 10% v/v HCl.

Dipping process
5. 3x3 cm^2 TiPt felt was dipped into the dip coating solution for 5 min.
6. The dipped TiPt felt was then dried at 100C for 10min in oven and followed by calcination at 500C for 10min in muffle furnace.
7. After that, the calcined TiPt felt was cooled down and weighed.
8. Repeat step 5-7 for several times until obtaining desired mass loading.

Remark: IrO_x mass loading can be raised by either increasing concentration of dip coating solution or dipping process.

Preparation of the Ni-2D-NC cathode electrode via spray coating method
The as-prepared Ni-2D-NC catalyst was spray-coated onto Sigracet 39BB carbon paper. Briefly, 50 mg of the Ni-2D-NC and 127µL or 10% by catalyst weight of XA-9 binder were ultrasonically mixed with 10ml of IPA for 60 min to form a dark colloidal solution which was then sprayed onto Sigracet 39BB carbon paper with 0.5 ml/min on 95 °C hotplate to obtain ~ 0.6 ± 0.1 mg/cm^2. To obtain 1.2 and 1.8 ± 0.1 mg/cm^2, 100 and 150 mg of Ni-2D-NC catalyst was added into the solution.

Electrochemical CO2RR measurements
A 5 cm^2 MEA electrolyzer was used to evaluate the CO2RR performance under modulated operating conditions. The Ni foam, AEM (X37-50 Grade RT, Dioxide Materials), and Ni-2D-NC cathode catalyst were employed as the anode, membrane, and cathode electrode, respectively. 50 ml of 0.5M KOH was recirculated and used throughout the experiment as the anolyte at 8 ml/min. Various CO_2 partial pressures (0.15, 0.3, 0.5, and 1.0) was introduced into the cathodic inlet under CO_2 flow rates of 15, 30, 50, and 100 ml/min (balancing by N_2 gas), respectively. The cathodic outlet was directly injected into an online gas-chromatography for gas products analysis.

3. **Results**

Influence of Ni-2D-NC mass loading on CO2RR performance under higher current densities

To solidify the impact of catalyst mass loading (m/l), we performed CO2RR using a Ni-2D-NC cathode catalyst with loadings of 0.6 and 1.2 mg/cm² across an applied current density range of 200 to 400 mA/cm². As shown in Fig. 1, at 200 to 300 mA/cm², the m/l of Ni-2D-NC had minimal influence on CO2RR performance, with FE(CO) remaining above 90%. However, a significant effect of catalyst m/l was observed at higher applied current densities. Increasing the current density from 300 to 350 mA/cm² caused the FE(CO) for 0.6 and 1.2 mg/cm² to abruptly drop to 76% and 82%, respectively. Further increasing the current density to 400 mA/cm² drastically decreased FE(CO) to around 28%, as HER effectively outcompeted CO2RR.

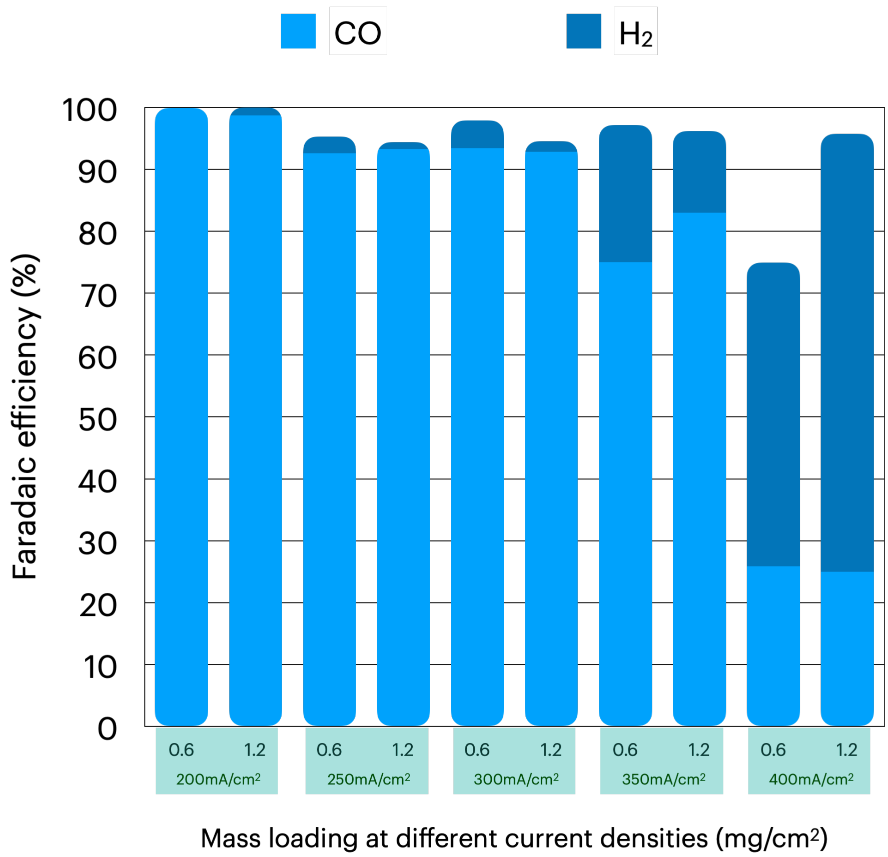

Figure 1. Faradaic efficiency of CO and H_2 of Ni-2D-NC catalyst with m/l of 0.6 and 1.2 mg/cm^2 under various applied current densities.

CO2RR performance of Ni-2D-NC cathode electrode under various catalyst mass loadings (m/l) and CO_2 partial pressures (PCO_2)

To identify the factors determining the CO2RR performance of the Ni-2D-NC cathode catalyst at different CO_2/N_2 mixture concentrations, we applied various CO_2 partial pressures of 0.15, 0.3, 0.5, and 1.0 at a constant total flow rate of 100 ml/min and an applied current density of 200 mA/cm^2. Regardless of the catalyst m/l, the CO2RR performance of Ni-2D-NC, whose

FE(CO) marginally increased with increasing $P(CO_2)$, was strongly impacted by CO_2 concentration.

To exclude the possibility that the poor performance of the Ni-2D-NC catalyst at low $P(CO_2)$ resulted from electrode optimization, various Ni-2D-NC mass loadings (0.6, 1.2, and 1.8 mg/cm^2) were prepared and tested. At $P(CO_2)$ of 0.15 (15% CO_2/85% N_2 mixture), the FE(CO) of the $Ni_{1.2\text{-}1.8}$-2D-NC catalyst was slightly higher than that of Ni0.6-2D-NC. Interestingly, we observed a remarkable increase in FE(CO) of $Ni_{1.2\text{-}1.8}$-2D-NC at higher $P(CO_2)$ compared to $Ni_{0.6}$-2D-NC, implying the importance of catalyst loading amount on CO2RR performance.

Additionally, the pulse electrochemical method was applied to study the improvement of Ni-2D-NC performance under a diluted 15% CO_2 condition to see if this could mitigate salt buildup and competitive HER. Preliminary results of pulse electrolysis under operation and regeneration potentials of 2.8V and 0V exhibited a substantial increase in FE(CO) to almost 80% with the $Ni_{1.2}$-2D-NC catalyst. Further verification of the pulse electrolysis operation is needed.

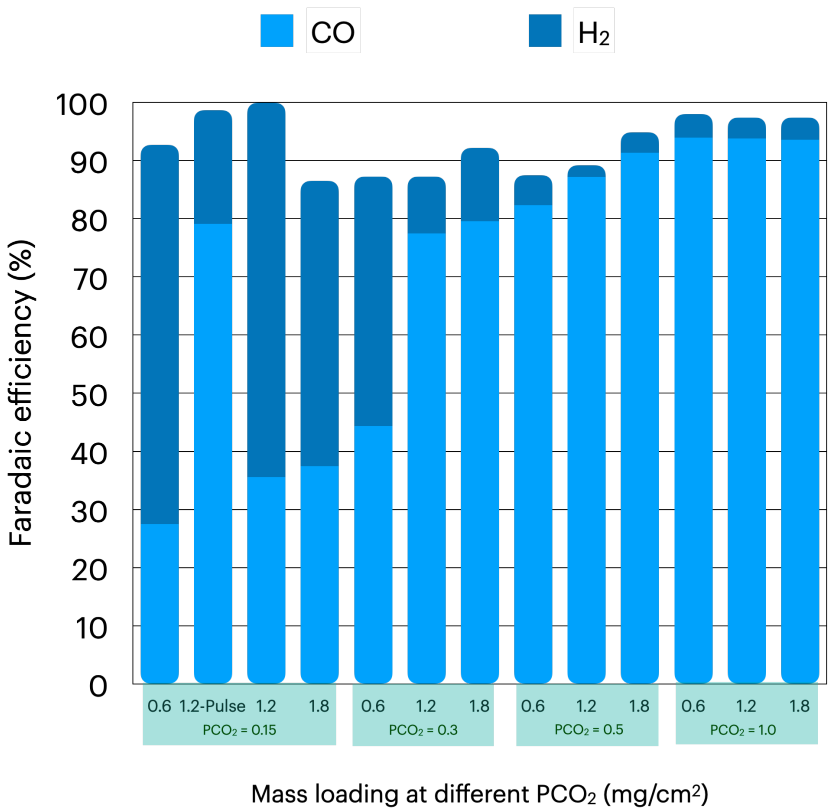

Figure 2. Faradaic efficiency of CO and H_2 of Ni-2D-NC cathode catalyst under different m/l (0.6 – 1.8 mg/cm^2) and PCO_2 (0.15 -1.0) at $200mA/cm^2$.

Studies of the possibility of using deposited IrO_x on TiPt felt as an anode

The IrO_x/TiPt-felt anode was prepared using different deposition methods (electrodeposition and thermal deposition) and varying loading amounts of IrO_x. Cyclic voltammetry (CV) profiles of the samples indicated that the presence of IrO_x on the TiPt felt enhanced the electrode's catalytic activity. IrO_x/TiPt-felt samples prepared by thermal deposition outperformed the others, with the thermal-deposited IrO_x/TiPt-felt having an IrOx m/l of 0.85 mg/cm^2 exhibiting the highest catalytic activity. Subsequently, thermal-deposited IrO_x/TiPt-felt samples with m/l values of 0.5 and 0.85 mg/cm^2 were deployed as the anode for CO_2 reduction reaction (CO2RR) performance testing. This was done in conjunction with a $Ni_{1.2}$-2D-NC cathode, under pure CO_2, in a 0.1M $KHCO_3$ electrolyte, and at a current density of 200 mA/cm^2. Chronopotentiometry (CP) profiles (Figure 3) illustrated that a higher IrO_x m/l anode electrode (0.85 mg/cm$_2$) performed CO2RR at a notably lower cell potential of ~3.4V compared to a lower m/l anode (~4.3V). Based on the CV profiles, it is worth noting that the IrO_x/TiPt-felt with an m/l of 1.5 mg/cm^2 was assumed to share similar results with the 0.5 mg/cm^2 sample based on their CV profiles.

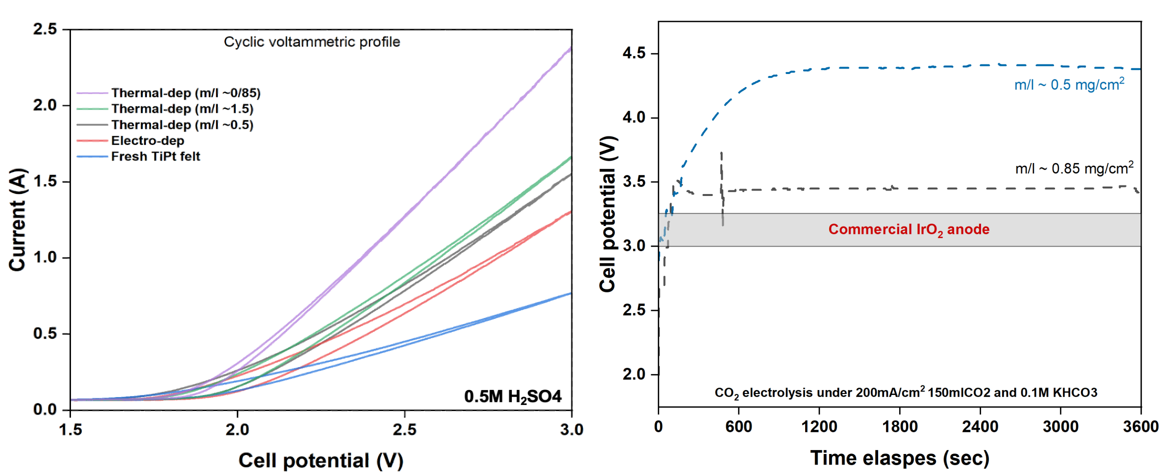

Figure 3. CV and CP profiles of various IrO_x/TiPt-felt samples.

4. Summary

In summary, for the Ni-2D-NC cathode, higher mass loadings were found to be beneficial at high current densities and low CO_2 concentrations. At 350 mA/cm^2, FE(CO) dropped to 76% and 82% for 0.6 and 1.2 mg/cm^2 loadings, respectively, and further decreased to around 28% at 400 mA/cm^2. At higher CO_2 partial pressures, the FE(CO) for Ni1.2-1.8-2D-NC significantly increased compared to Ni0.6-2D-NC, underscoring the importance of catalyst loading. Under diluted 15% CO_2 conditions, applying a pulse electrochemical method significantly improved the CO2RR performance of $Ni_{1.2}$-2D-NC with nearly 80% of FE(CO), indicating substantial mitigation of salt formation and competitive HER. CV and CP profiles demonstrated that a thermal-deposited IrO_x/TiPt-felt with an m/l of 0.85mg/cm^2 exhibited the highest catalytic activity at a significantly lower cell potential.

5. Upcoming tasks

- Improvement of CO2RR performance of $Ni_{1.8mgcm-2}$-2D-NC catalyst under PCO_2 of 0.15 using pulse electrochemical method:
  - Influence of operation and regeneration times
  - Influence of applied pulse potential
- Investigation of CO2RR performance of additional 5mg/cm^2 CALF20 on the GDE as a surface-accessible CO_2 layer under a diluted CO2 condition (PCO_2 of 0.15).